BIOI621 - Genome Assembly and Annotation

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Gene Expression Analysis of MMP2 and MMP9 in Response to Scaffold Treatment

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Abstract

Macrophages are key regulators of tissue repair and biomaterial integration, partly through the activity of matrix metalloproteinases (MMPs) such as MMP-2 and MMP-9, which degrade extracellular matrix components during inflammation and remodeling. Zinc-based calcium phosphate (CaP) scaffolds are widely studied for bone regeneration, and ion modifications with elements such as manganese (Mn) and lithium (Li) have been proposed to enhance immune modulation. This research investigated whether macrophage expression of MMP-2 and MMP-9 varies in response to zinc-based CaP scaffolds with ion modifications. We analyzed publicly available RNA-seq data (GSE224524) from RAW264.7 macrophages cultured on three scaffold types: CaP Zn, CaP Zn0.8Mn, and CaP Zn0.8Mn0.1Li. The reads were processed using HISAT2 for alignment, StringTie for assembly and quantification, and R for statistical analysis. MMP-2 expression was undetectable across all treatments. MMP-9 showed variable expression, with the highest average in CaP Zn (12.6 FPKM); however, statistical testing indicated no significant differences among groups (Kruskal–Wallis $p = 0.37$). These findings suggest that ion modifications to CaP scaffolds did not significantly alter macrophage MMP expression under the tested conditions.

Introduction

Successful tissue repair and biomaterial integration depend heavily on the immune system, with macrophages as key regulators. Through their ability to adopt different activation states, macrophages influence inflammation, and extracellular matrix (ECM) remodeling. A central component of this remodeling is the activity of matrix metalloproteinases (MMPs), which are enzymes that degrade ECM proteins to enable wound healing and tissue regeneration. Among these, MMP-2 and MMP-9 are especially important: MMP-9 is strongly associated with inflammation and tissue turnover, while MMP-2 contributes to collagen degradation and vascular remodeling.

Calcium phosphate (CaP) scaffolds are widely used biomaterials in bone regeneration because of their biocompatibility and structural similarity to bone mineral. Recent studies suggest that incorporating zinc (Zn) and other ions such as manganese (Mn) and lithium (Li) into CaP scaffolds can further enhance scaffold performance by modulating macrophage polarization toward an anti-inflammatory, pro-regenerative phenotype. While much of this research has focused on macrophage polarization markers (e.g., M1/M2 signatures), less is known about how ion-modified scaffolds influence MMP expression, which directly mediates ECM turnover during tissue integration.

In this research, we analyzed RNA-seq data from RAW264.7 macrophages cultured on CaP scaffolds with different ion modifications (Zn, Zn0.8Mn, Zn0.8Mn0.1Li) to investigate whether the scaffold alters MMP-2 and MMP-9 expression. By integrating alignment, assembly, and gene expression analysis tools, we aimed to determine whether these scaffold modifications significantly impact macrophage-mediated ECM remodeling pathways.

Materials & Method

Data Source

RNA-seq data were obtained from the NCBI Gene Expression Omnibus (accession GSE224524). The dataset consisted of RAW264.7 macrophages cultured on three scaffold conditions:

zinc-based calcium phosphate (CaP Zn), zinc–manganese alloy scaffold (CaP Zn0.8Mn), and zinc–manganese–lithium alloy scaffold (CaP Zn0.8Mn0.1Li), with two biological replicates per treatment group.

Data Acquisition and Quality Control

Raw FASTQ files were downloaded from the Sequence Read Archive using the SRA Toolkit (fasterq-dump). Sequence quality was assessed with FastQC, and summary reports were generated with MultiQC to aggregate results across all samples. All six samples displayed consistently high sequence quality with mean Phred scores greater than 30, minimal adapter contamination, and no need for trimming.

Read Alignment

Sequencing reads were aligned to the Mus musculus reference genome (mm10) using HISAT2. The resulting SAM files were converted to BAM format, sorted, and indexed with SAMtools. Alignment statistics, including overall mapping rates, were obtained from the HISAT2 summary log files.

Transcript Assembly and Quantification

Transcript assembly and quantification were performed using StringTie (v2.x) with the GENCODE vM37 annotation to enable reference-guided assembly. For each sample, StringTie produced gene abundance tables (*_gene_abund.tab) containing normalized expression values reported as fragments per kilobase of transcript per million mapped reads (FPKM).

Gene Expression Analysis

Expression data were imported into R and merged with sample metadata. Genes of interest (Mmp2 and Mmp9) were extracted for downstream analysis. Gene expression levels were summarized as mean FPKM values per treatment group and visualized using ggplot2. Bar plots were generated to display average expression across treatments, while dot plots illustrated replicate-level variation.

Statistical Analysis

Statistical testing was performed in R. Normality of residuals was assessed with the Shapiro–Wilk test, and homogeneity of variance was evaluated with Levene's test. Because the variance assumption was violated, the Kruskal–Wallis test was used to compare Mmp9 expression across treatments. Mmp2 expression was uniformly zero across all samples, which led us to having no statistical analysis for Mmp2.

Result

Quality Control

The raw sequencing reads from all six samples exhibited consistently high quality, as assessed with FastQC and summarized using MultiQC. Phred scores remained above 30 across all base positions, with an average of approximately 35 *(Figure 1).* This corresponds to a base-calling accuracy greater than 99%, indicating minimal sequencing errors. The GC content and sequence length distributions were within expected ranges for mouse transcriptome data. Adapter contamination was negligible, demonstrating that the sequencing libraries were well prepared. As a result, no adapter trimming or additional filtering was required prior to alignment.

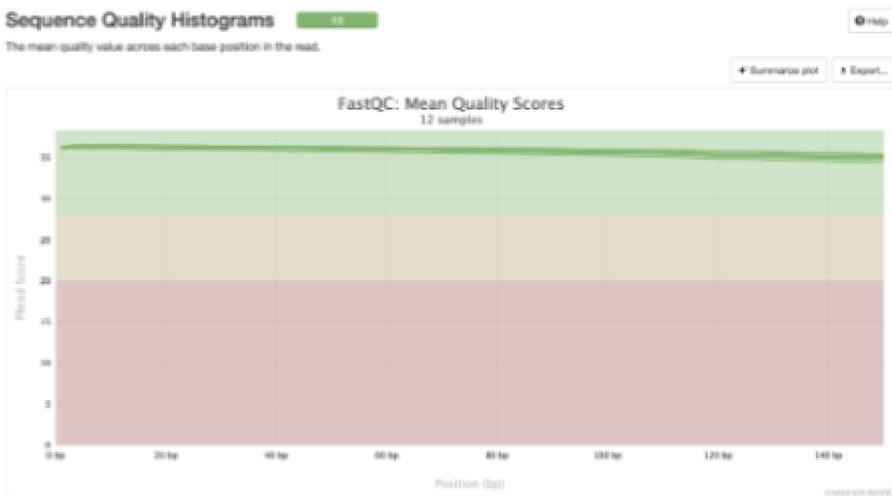

Figure 1. Sequence quality assessment of RNA-seq data. MultiQC summary plot of FastQC mean quality scores across all 12 sequencing libraries (paired-end reads from six samples). Phred quality scores remained consistently above 30 across all base positions, corresponding to a base-calling accuracy greater than 99%.

Read alignment

Following quality assessment, sequencing reads were aligned to the *Mus musculus* (mm10) reference genome using HISAT2. All samples achieved high alignment rates, with over 90% of reads successfully mapped *(Figure 2)*. These results reflect both the high quality of the sequencing libraries and the suitability of the reference genome. The consistently strong mapping efficiency also suggests minimal contamination and provides a reliable foundation for downstream transcript assembly and expression analysis.

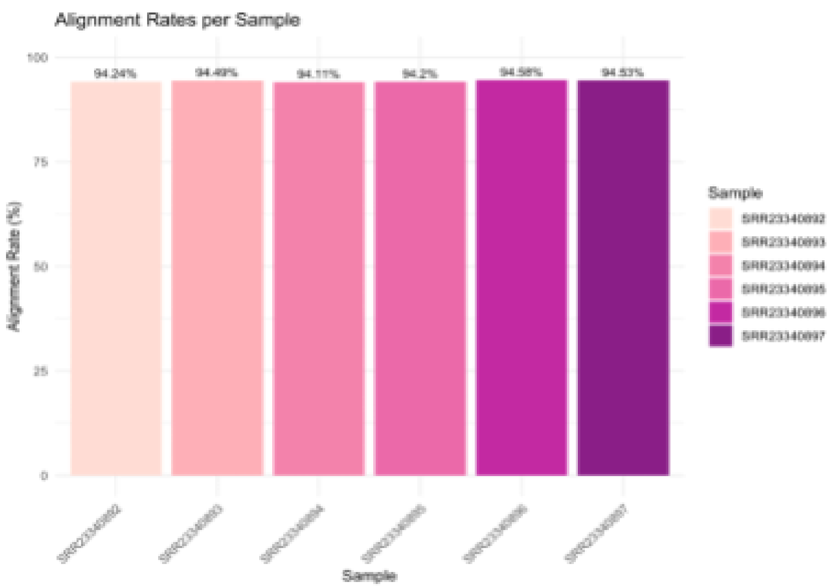

Figure 2. Alignment rates of RNA-seq samples. Bar plot showing overall alignment rates for each sample based on HISAT2 summary reports. All six samples achieved >90% alignment, indicating high-quality sequencing data and appropriate reference genome selection.

Transcript Assembly & Annotation

Aligned reads were processed with StringTie for transcript assembly and quantification. Using the GENCODE vM37 annotation file enabled reference-guided assembly, ensuring accurate transcript assignment. For each sample, StringTie generated transcript-level annotation files (*.gtf) and gene abundance tables (*.tab) containing normalized expression values (FPKM). These outputs provided the basis for downstream expression analysis of *Mmp2* and *Mmp9* across scaffold treatments.

MMP9 Expression

Expression of Mmp9 was detected across all scaffold conditions. Average FPKM values were highest in CaP Zn (12.6), followed closely by CaP Zn0.8Mn0.1Li (12.4), and lowest in CaP Zn0.8Mn (10.1) (Figure 3). The dot plot (Figure 4) illustrates replicate-level variation, with CaP

Zn showing consistently higher values, while CaP Zn0.8Mn replicates displayed greater variability.

Statistical analysis with the Kruskal–Wallis test revealed no significant differences in Mmp9 expression between scaffold treatments (chi-squared = 2.00, p = 0.37). These findings suggest that scaffold ion modifications with Mn or Li did not significantly alter Mmp9 expression in macrophages under the tested conditions.

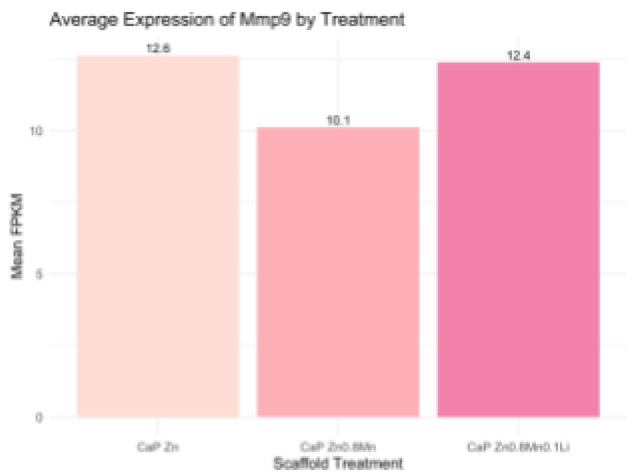

Figure 3. Average expression of Mmp9 across scaffold treatments. Bar plot showing mean FPKM values for *Mmp9* in CaP Zn, CaP Zn0.8Mn, and CaP Zn0.8Mn0.1Li.

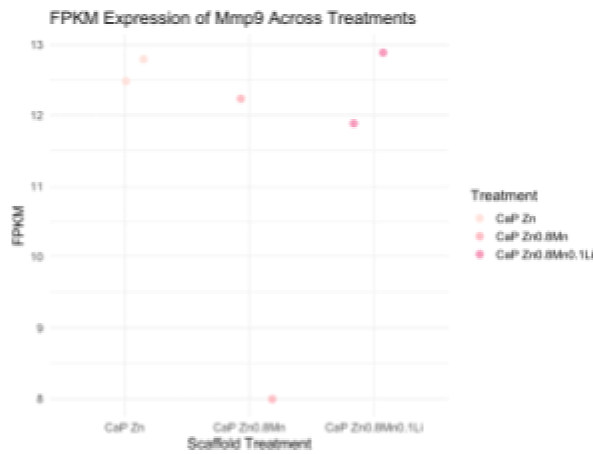

Figure 4: Replicate-level variation in Mmp9 expression. Dot plot showing individual sample FPKM values by treatment group.

MMP2 Expression

In contrast to Mmp9, Mmp2 expression was undetectable across all scaffold treatments. Both the bar plot (Figure 5) and dot plot (Figure 6) show uniform values of zero FPKM for every replicate. Because expression was consistently absent across groups, statistical analysis could not be performed. These results indicate that Mmp2 was not measurably expressed in RAW264.7 macrophages under any scaffold condition.

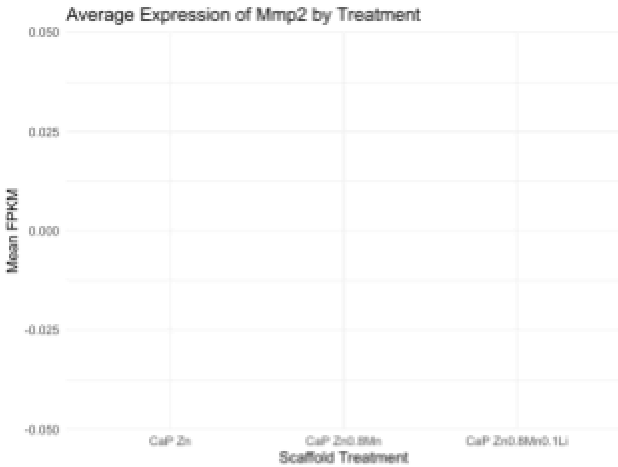

Figure 5. Average expression of Mmp2 across scaffold treatments. Bar plot showing mean FPKM values for *Mmp2* in CaP Zn, CaP Zn0.8Mn, and CaP Zn0.8Mn0.1Li. Expression was uniformly zero across all treatments.

Figure 6. Replicate-level variation in Mmp2 expression. Dot plot showing FPKM values for individual replicates. Expression was uniformly zero across all samples.

Discussion

This research investigated whether zinc-based calcium phosphate scaffolds with ion modifications influence macrophage expression of MMP-2 and MMP-9. Our analysis revealed that MMP-2 was undetectable under all scaffold conditions, while MMP-9 was expressed at varying levels. Among treatments, CaP Zn showed the highest average MMP-9 expression (12.6 FPKM), followed by CaP Zn0.8Mn0.1Li (12.4 FPKM), and CaP Zn0.8Mn (10.1 FPKM). However, statistical testing using the Kruskal–Wallis test indicated that these differences were not significant (Chi-squared = 2.00, $p = 0.37$).

These findings suggest that scaffold ion modifications with Mn and Li do not significantly alter macrophage MMP expression in this model. While MMPs are well-established mediators of extracellular matrix degradation and remodeling, our results indicate that they may not be the primary pathway through which these scaffolds modulate macrophage behavior. Instead, scaffold effects may be more evident in macrophage polarization markers or cytokine production, as suggested by Qiang et al. (2023), who reported that Zn-Mn-Li scaffolds promoted anti-inflammatory (M2) macrophage phenotypes. Similarly, Wissing et al. (2019) emphasized the role of scaffold microarchitecture in macrophage-driven degradation, highlighting the multifactorial nature of scaffold–immune interactions.

Our research has several limitations. The dataset included only two replicates per group, limiting statistical power. Because the data were derived from a public repository, experimental variables such as scaffold preparation and macrophage culture conditions could not be controlled. Future studies should include larger sample sizes and additional biological replicates, along with expanded analyses of cytokines, other MMP family members, and tissue inhibitors of metalloproteinases (TIMPs). Time-course experiments may also help determine whether scaffold effects on MMP expression are transient or delayed.

In summary, our results indicate that zinc-based CaP scaffold modifications with Mn and Li did not significantly affect macrophage MMP-2 or MMP-9 expression. Although CaP Zn exhibited the highest average MMP-9 expression, the difference was not statistically significant. These findings suggest that macrophage responses to ion-modified scaffolds may occur through alternative molecular pathways, underscoring the importance of integrative approaches to studying biomaterial–immune interactions.

Conclusion

Our research examined whether zinc-based calcium phosphate scaffolds with ion modifications influence macrophage expression of MMP-2 and MMP-9. Using RNA-seq data from RAW264.7 macrophages, we found that MMP-2 expression was undetectable across all scaffold conditions, while MMP-9 was expressed but showed no statistically significant differences among treatments. Although CaP Zn exhibited the highest average MMP-9 expression, scaffold modifications with manganese and lithium did not significantly alter expression levels. These results suggest that macrophage responses to ion-modified scaffolds may be mediated through pathways other than MMP activity. Future work incorporating larger sample sizes, time-course analyses, and additional markers of immune modulation will be important for clarifying how scaffold composition influences macrophage behavior and tissue integration.

Contributions:

Manuela Deigh: Found and Downloaded the RNA-seq dataset and performed quality control analyses. Conducted read alignment using HISAT2 and SAMtools, followed by transcriptome assembly and annotation with StringTie. Generated figures for alignment rates, *Mmp9* and *Mmp2* bar plots, and *Mmp9* and *Mmp2* dot plots. Performed statistical testing on expression data. Drafted the PowerPoint presentation and contributed to the final report.

Adetomiwa Adeusi: Worked on data preprocessing and pipeline setup. Did read alignment and verification of HISAT2 and SAMtools outputs, and performed a backup alignment using STAR. Helped with transcriptome assembly and annotation. Reviewed and interpreted the gene expression results for *Mmp9* and *Mmp2*. Drafted the final report and peer reviewed the PowerPoint presentation.

References

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